



Liquid Flow Meter using Microcontroller

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BACHELOR OF ENGINEERING  
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO  
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF  
  
THE REQUIREMENTS FOR THE DEGREE OF  
BACHELOR OF ENGINEERING  
IN COMPUTER ENGINEERING

2023

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# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 Background and Rationale**

This project was created to facilitate the irrigation of trees. Currently, watering plants requires walking to pick up a hose to spray water or using a watering can to pour water, which is a waste of time. So our group invented the Liquid Flow Meter using a Microcontroller. To save time make it possible to reap benefits in your life.

### **1.2 Objective**

1.2.1 Flow Measurement: The main objective is to accurately measure the flow of liquids. It involves determining the volume or mass of liquid flowing through a flow meter per unit of time. This information is important in various industrial, commercial or research applications.

1.2.2 The flow meter must provide accurate measurements. To ensure reliable information This is important for processes where precise control of fluid flow is important, such as in chemical processing, water management or industrial manufacturing.

### **1.3 Scope**

1.3.1 General User

1.3.2 Automatically use programmed rules and mechanical concepts

1.3.3 Design and dedicate this machine.

1.3.4 Eliminate the problem of travelling to other provinces. Don't worry about finding someone to take care of the garden.

### 1.3.5 Hardware

- Water flow sensor
- Arduino R3
- LCD
- Connector with internal threading
- Connecting wires
- Laptop

## 1.4 Methodology

### 1.4.1 Proposal preparation and defense

- Identify the scope of the project
- Draft and sketch of scope the project

### 1.4.2 Conceptual system design

- Decision for this project

### 1.4.3 Detailed system design

- Design an image of the product
- Design hardware and software will be used

### 1.4.4 Computer Engineering Pre-Project Document Preparation

- Documentation introduction
- Documentation literature review
- Document design

### 1.4.5 Computer Engineering Project EXAM

#### 1.4.6 System development

- Hardware
- Software

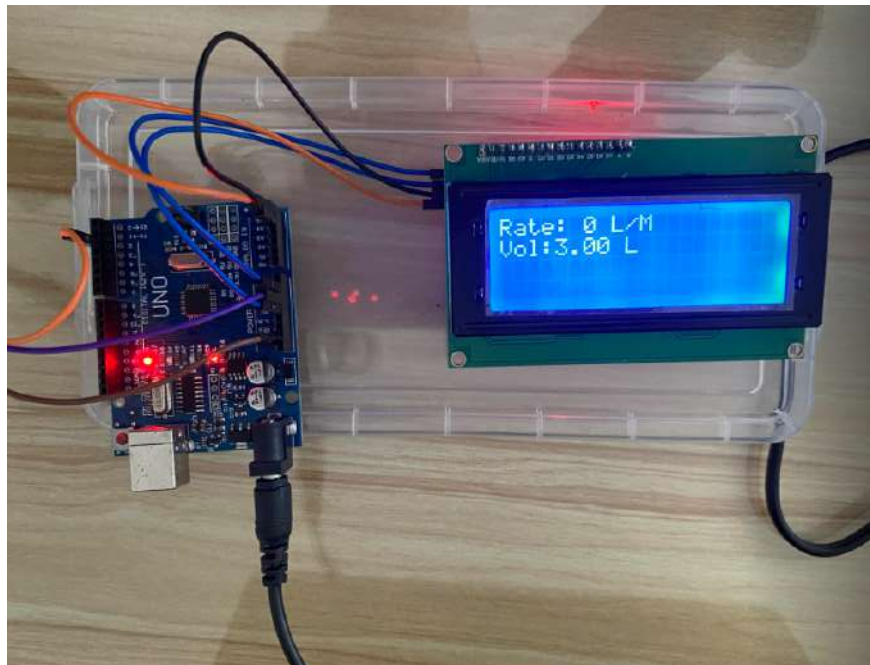
#### 1.4.7 System testing

- Testing Hardware Systems

#### 1.4.8 Computer Engineering Pre-Project 2 exam

### 1.5 Circuit

- Circuit of liquid flow meter using microcontroller



## 1.6 Plan

### Table 1.1 Working plan

[illegible]

## 1.7 Budget

| No    | Materials/Equipment               | Amount | Price(Baht) |
|-------|-----------------------------------|--------|-------------|
| 1     | Water flow YFS201                 | 1      | 148         |
| 2     | LCD                               | 1      | Free        |
| 3     | Arduino R3                        | 1      | Free        |
| 4     | Connector with internal threading | 2      | Free        |
| 5     | Connecting wires                  | 10     | Free        |
| 6     | Pipe 5 meter                      | 1      | 250         |
| 7     | Mini tank                         | 1      | 189         |
| Total |                                   |        | 587         |

This is an estimation of the equipment that we use in this project based on the current online shopping site.

# **CHAPTER 2**

## **LITERATURE REVIEW**

### **2.1 Introduction**

Water is essential on Earth, but due to poor resource management, water resources are decreasing every day. Industries, for the most part, are involved in managing water resources, and it is necessary to have devices like water flow sensors to monitor water usage accurately. The water flow sensor plays a crucial role in measuring and controlling the flow of liquids precisely, especially in situations that require efficient water management and real-time display of data. To enhance efficiency in process control and contribute to technological advancements, utilizing accurate flow data allows us to respond effectively to water management needs and fluid control in various industries. This includes agriculture, food processing, manufacturing, mining, water recycling, irrigation, and other sectors. These sensors enable us to use water valuably, similar to how we value the precious resources of our planet.

### **2.2 Related Technology**

**2.2.1 Flow Sensor:** A flow sensor is a device that measures the rate of liquid flow. There are various types of flow sensors, including turbine, electromagnetic, ultrasonic, and thermal sensors. The chosen sensor type depends on the specific requirements of the application.

**2.2.2 Microcontroller:** The microcontroller serves as the brain of the system. It processes information from the flow sensor, executes programmed instructions, and makes decisions based on the input received. Common microcontrollers include those from the Arduino or Raspberry Pi families.

**2.2.3 Programming:** The microcontroller is programmed to interpret the signals from the flow sensor and execute tasks accordingly. The programming logic can include tasks such as displaying flow rates, logging data, or triggering actions based on certain flow conditions.

2.2.4 User Interface (Optional): In some applications, a user interface may be incorporated to allow interaction with the system. This could be a simple display, buttons, or a more sophisticated graphical user interface (GUI) depending on the complexity of the setup.

## 2.3 Critterai Compari

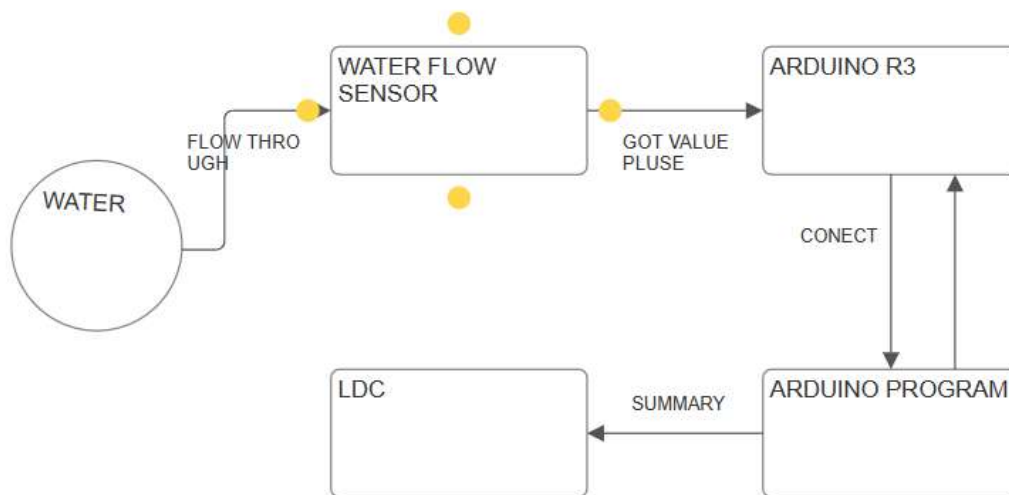
| Diagram                                                                             | High price | Mobile control | Save electricity | Easy to use | convenience | Sources                                                |
|-------------------------------------------------------------------------------------|------------|----------------|------------------|-------------|-------------|--------------------------------------------------------|
|   | ✓          | ✓              | ✓                | ✓           | ✓           | <a href="http://www.shopee.co.th">www.shopee.co.th</a> |
|  | ✓          | ✗              | ✓                | ✓           | ✓           | <a href="http://www.shopee.co.th">www.shopee.co.th</a> |
|  | ✗          | ✗              | ✓                | ✓           | ✓           | From my project                                        |

# CHAPTER 3

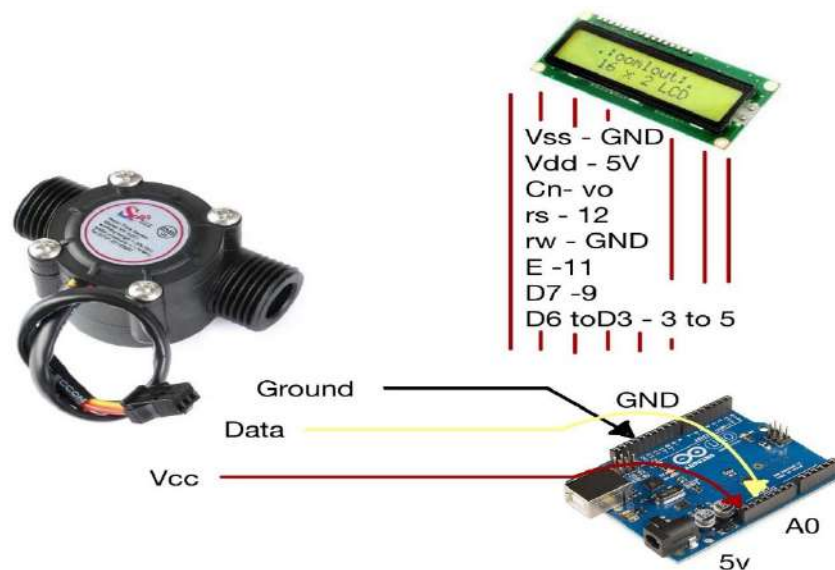
## ANALYSIS AND DESIGN

### 3.1 Block Diagram

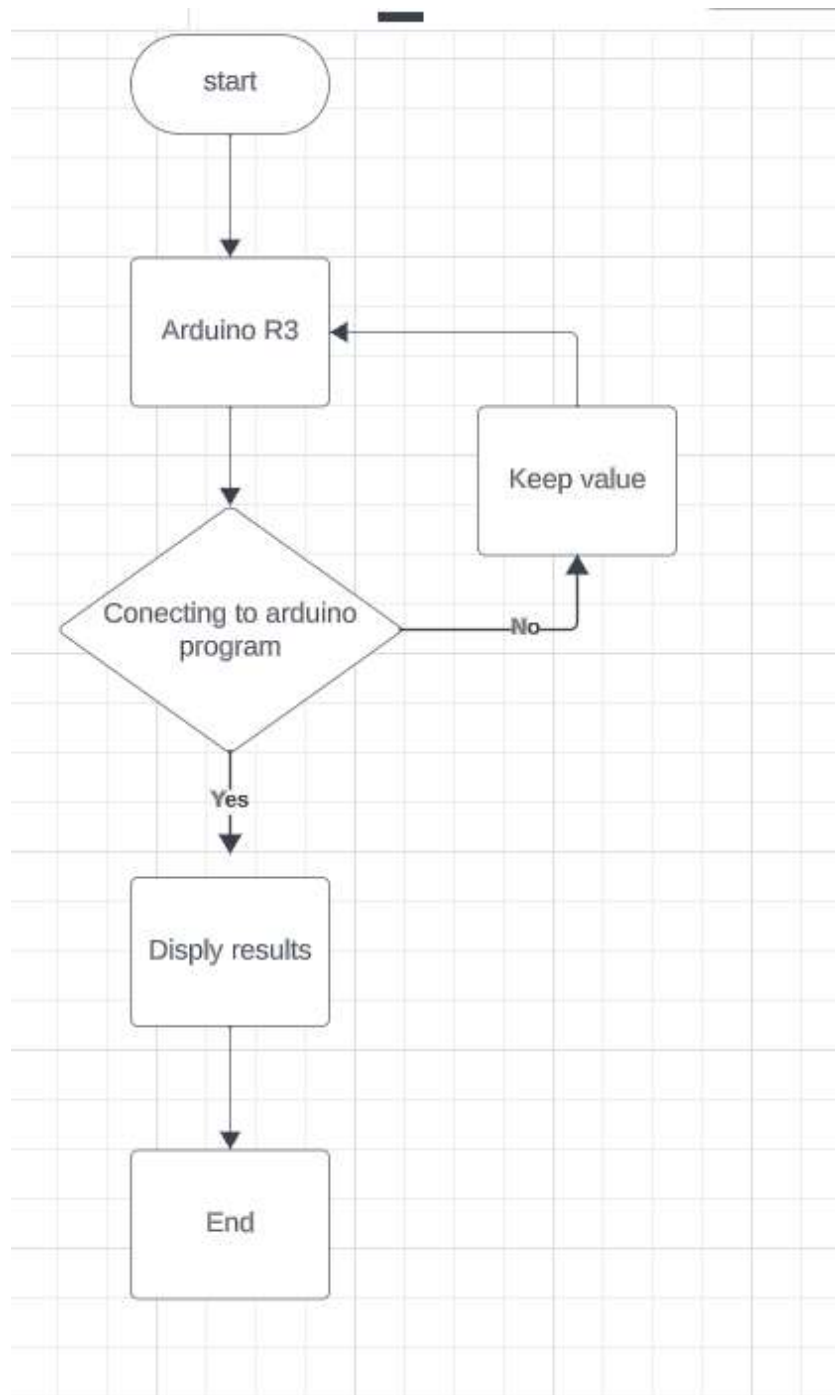
Sensor Water flow is a Sensor that measures the flow rate of liquid passing through the Sensor. It can be used to measure liquids. The signal received from the Sensor will come out as a Pulse. If you get the pulse value, write an Arduino program to receive the Interrupt value.



### 3.2 Circuit Diagram



### 3.3 Flow Chart



### 3.4 Arduino water flow sensor code

```
volatile int flow_frequency; // Measures flow sensor pulses

// Calculated litres/hour

float vol = 0.0, l_minute;

unsigned char flowsensor = 2; // Sensor Input

unsigned long currentTime;

unsigned long cloopTime;

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

void flow () // Interrupt function

{

    flow_frequency++;

}

void setup()

{

    pinMode(flowsensor, INPUT);

    digitalWrite(flowsensor, HIGH); // Optional Internal Pull-Up

    Serial.begin(9600);

    lcd.begin();
```

```

attachInterrupt(digitalPinToInterrupt(flowsensor), flow, RISING); // Setup InterruptX

lcd.clear();

lcd.setCursor(0,0);

lcd.print("Water Flow Meter");

lcd.setCursor(0,1);

lcd.print("Circuit Digest");
|
currentTime = millis();

cloopTime = currentTime;

}

```

```

void loop ()

{

    currentTime = millis();

    // Every second, calculate and print litres/hour

    if(currentTime >= (cloopTime + 1000))

    {

        cloopTime = currentTime; // Updates cloopTime

        if(flow_frequency != 0){

            // Pulse frequency (Hz) = 7.5Q, Q is flow rate in L/min.

            l_minute = (flow_frequency / 7.5); // (Pulse frequency x 60 min) / 7.5Q = flowrate in L/hour

            lcd.clear();

            lcd.setCursor(0,0);

            lcd.print("Rate: ");

            lcd.print(l_minute);

            lcd.print(" L/M");

            l_minute = l_minute/60;

            lcd.setCursor(0,1);

```

```

    lcd.print("Vol:");

    lcd.print(vol);

    lcd.print(" L");

    flow_frequency = 0; // Reset Counter

    Serial.print(1_minute, DEC); // Print litres/hour

    Serial.println(" L/Sec");

}

else {

    Serial.println(" flow rate = 0 ");

    lcd.clear();

    lcd.setCursor(0,0);

    lcd.print("Rate: ");

    lcd.print( flow_frequency );

    lcd.print(" L/M");

    lcd.setCursor(0,1);

    lcd.print("Vol:");

    lcd.print(vol);

    lcd.print(" L");

}

}

}

```

### 3.5 Water flow sensor flow rate

In general, when we using water at a water pressure of 1 bar or equal to normal water pressure at home through a ½ inch faucet. The water volume will be 30 liters when turn on for 1 minute

#### Equation of velocity in water pipes from Newton's equation

$$V = Q/A$$

V = velocity, Q = water flow rate, A= cross-sectional area of water pipe

**FROM** water flow rate  $5 \times 10^{-4} \text{ m}^3/\text{s}$

cross-sectional area of water pipe  $9.428 \times 10^{-5} \text{ m}^2$

$$V = \frac{5 \times 10^{-4}}{9.428 \times 10^{-5}}$$

$$V = 5.303 \text{ m/s}$$

∴ According to calculations, when water flows through the pipe at a speed lower than 5.303 meters per second. It's will be cause distorted results.

### 3.6 Oscilloscope

Oscilloscopes are not typically used directly with liquid flow meters, but they can be used in the context of electronics and signal processing involving certain types of flow sensors.

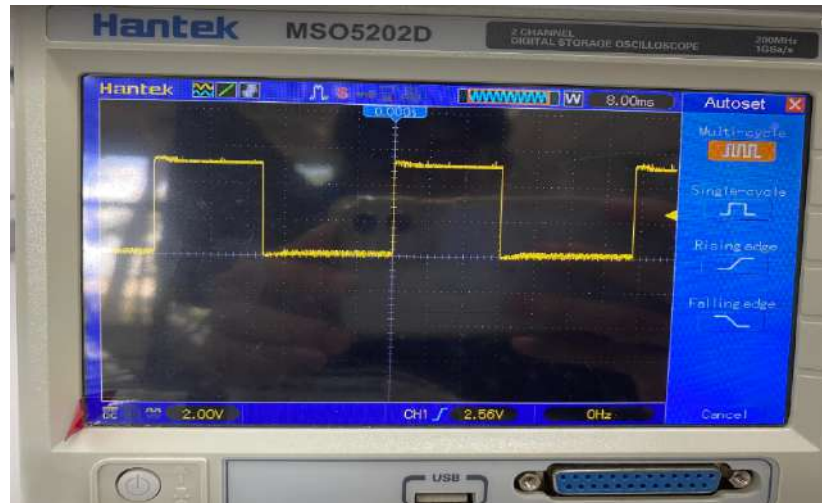


Figure1: Graph of liquid flow meter

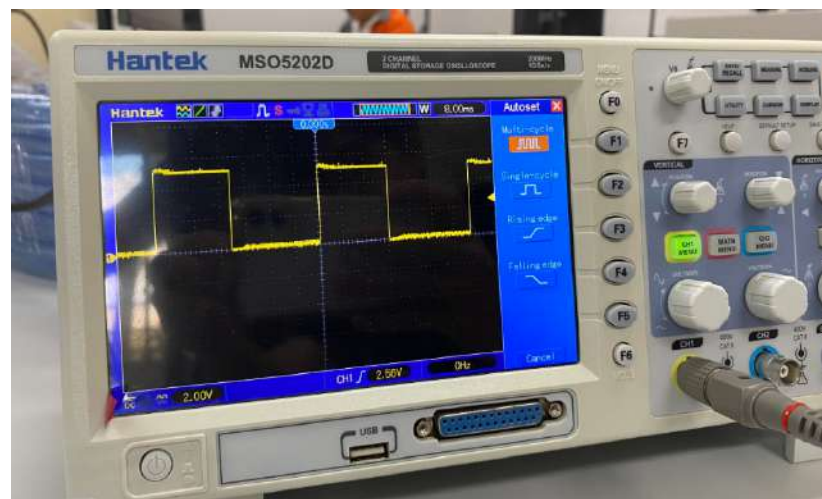


Figure2: Graph of liquid flow meter

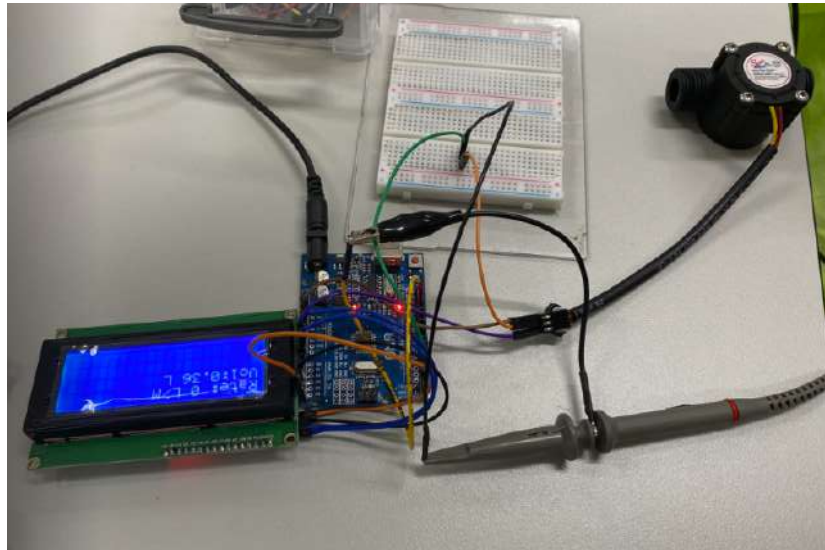


Figure3 : Connecting the circuit to the Oscilloscope

From the analysis of the signal in Figure 1 and Figure 2, which is used to measure the water flow rate, it is evident that the water flows relatively consistently without significant oscillations. This results in high accuracy in the data we obtain.

### 3.7 Liquid flow meter using microcontroller

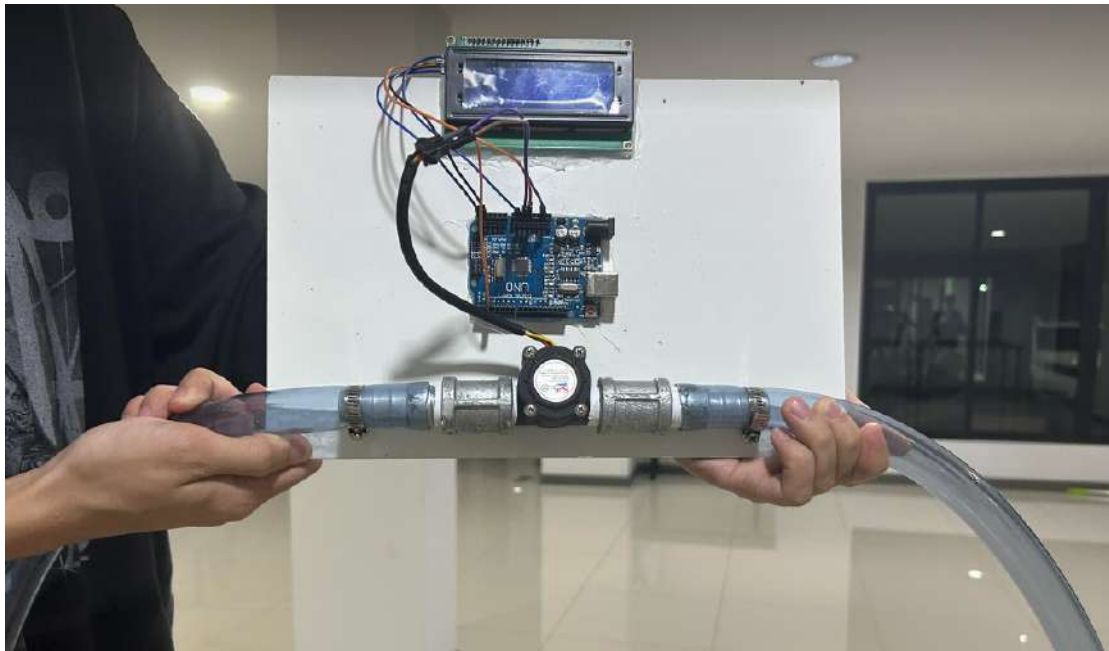


Figure 1

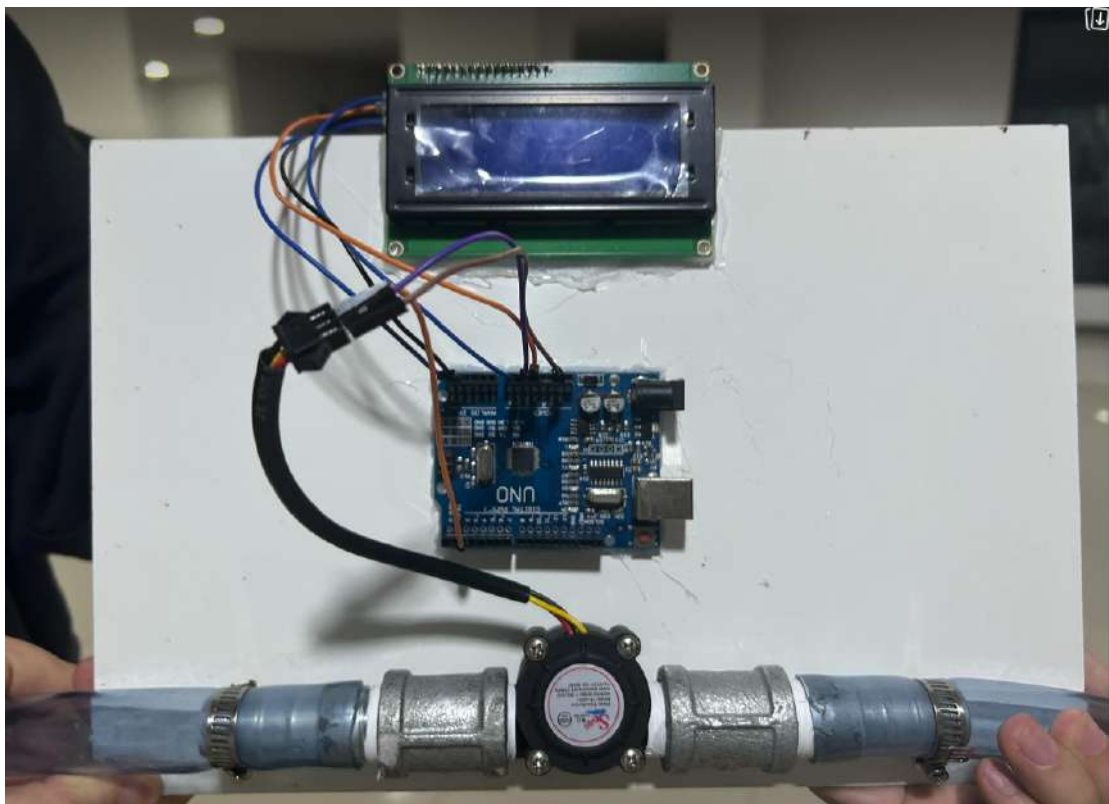


Figure 2

### 3.8 Experimental results

| LIQUID FLOW WITH MICROCONTROLLER |                                                                                     |                                                                                     |                                                                   |                                                              |
|----------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|
| NO                               | WATER VOLUME<br>2 LITERS                                                            | WATER VOLUME<br>4 LITERS                                                            | Reasoning                                                         |                                                              |
|                                  |                                                                                     |                                                                                     | 2 LITERS                                                          | 4 LITERS                                                     |
| 1                                |    |    | The cause of the error was that the hose was adjusted too deeply. | This is due to an incorrect measurement of the water volume. |
| 2                                |   |   | The cause of the error was that the hose was adjusted too deeply. | This is due to an incorrect measurement of the water volume. |
| 3                                |  |  | Perfect                                                           | This is due to an incorrect measurement of the water volume. |
| 4                                |  |  | Perfect                                                           | This is due to an incorrect measurement of the water volume. |
| 5                                |  |  | Perfect                                                           | Perfect                                                      |

## **CHAPTER 4**

### **CHAPTER**

#### **4.1 Conclusion**

This project is created to MEASURE FLOWIND WATER. This device can measure how much water has flowed. This measurement our group uses in farming. To measure the appropriate amount of water and use the personnel as high as possible.

#### **4.2 Discussion**

Result of working in senior projects. Liquid flow meters with microcontrollers are quite accurate. But the level must on measured according to the machine's criteria, that is, there must be water used for measurement flowing through the machine at a speed greater than 5.4 meters per second. In comparison to the flow meters available in the market, this liquid flow meter is relatively inexpensive and offers performance comparable to market-standard measurement devices.

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